

LLM Cross-Sectional Rankings as Implicit Factor Models

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LLM CROSS-SECTIONAL RANKINGS AS IMPLICIT FACTOR MODELS

From Point-in-Time Data to Portfolios, Factors, and Diagnostics

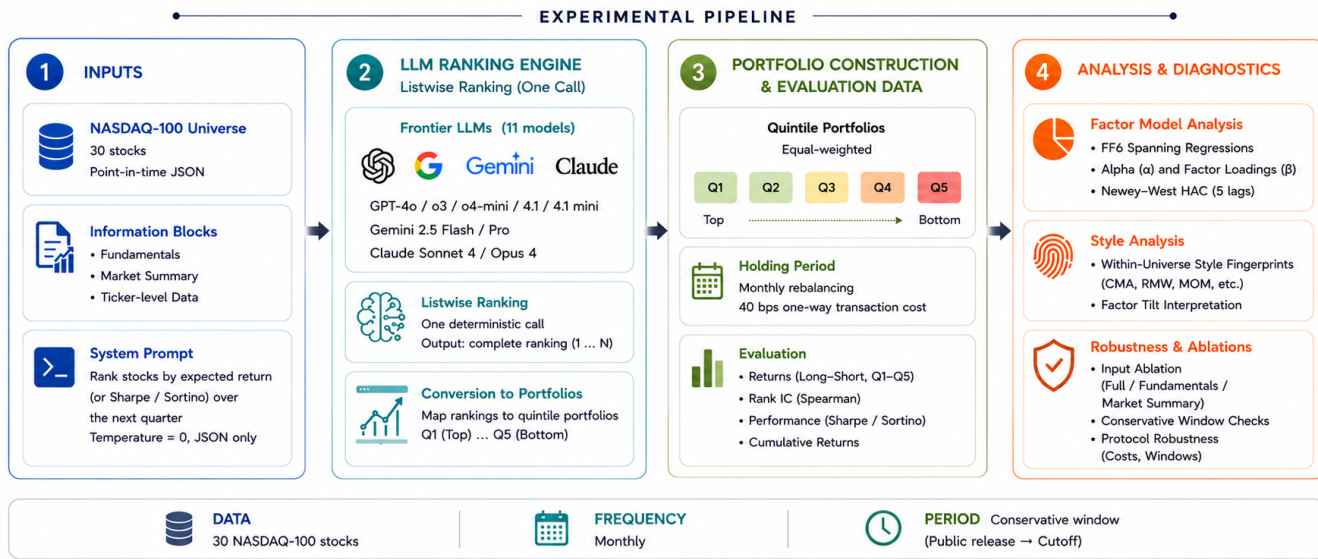


Figure 1: Experimental pipeline for deterministic listwise LLM ranking and portfolio, factor, ablation, and cost diagnostics.

ABSTRACT

Can frontier LLMs be evaluated as factor-like cross-sectional rankers rather than as point forecasters? We study 11 included LLMs from OpenAI, Google, and Anthropic on 30 NASDAQ-100 constituents using a controlled listwise ranking protocol, with expected return as the primary objective and Sharpe/Sortino prompts as robustness checks. Each monthly ranking is converted into quintile portfolios and evaluated with rank IC, FF6 spanning regressions, within-universe style fingerprints, common-window checks, and transaction costs. We find that (i) model choice materially changes the induced factor profile: in the conservative window, o3 reaches a long-short Sharpe of 5.54 and IC of 0.59, while GPT-4o variants produce weak or negative spreads; (ii) the strongest LLM portfolios in this benchmark carry large FF6 alphas with positive momentum and negative value/profitability/investment loadings, consistent with a growth-momentum implicit prior; and (iii) input ablations show that market summaries reproduce most of the full-input factor profile, whereas fundamentals-only prompts weaken both alpha and style coherence. The evidence is a controlled model-comparison benchmark rather than a production trading claim: choosing an LLM changes the implicit factor exposure being tested. An anonymous code and artifact repository is available at https://github.com/llm-CS-ranking/llm_CS_ranking.

CCS CONCEPTS

• **Computing methodologies** → **Learning to rank**; *Natural language generation*; • **Applied computing** → **Economics**.

KEYWORDS

large language models, stock ranking, listwise learning to rank, portfolio construction, factor analysis, quantitative finance

1 INTRODUCTION

Large language models have opened new possibilities for financial analysis: unlike traditional quantitative factors, they can synthesize earnings trends, competitive dynamics, and macro context into investment judgments with no task-specific training. The empirical question is *how* to formulate this capability as a reproducible financial task rather than an anecdotal stock-picking exercise. Prior work has relied either on coarse rating classification [34]—too coarse for portfolio construction and typically evaluated on a single model without backtest—or on raw return prediction, which is dominated by noise [15]. We argue that *ranking*—ordering stocks from most to least attractive—is the most natural middle ground: it is strictly more general than classification or regression, sidesteps the magnitude-noise of returns, and directly enables quintile portfolios, long-short strategies, and Spearman rank-IC evaluation. Our factor-model perspective treats each LLM as an implicit cross-sectional prior: given the same point-in-time information set, different models induce different expected-return orderings, style exposures, and residual

alphas. The object is therefore a controlled benchmark of model-induced priors, not a claim that a short-window NASDAQ backtest is directly deployable as a production strategy.

We organize the study around four research questions: **RQ1**: do LLM rankings embed distinct factor-like priors beyond classical FF6 premia? **RQ2**: how do factor loadings and alphas change when the LLM sees fundamentals only, market summaries only, or both? **RQ3**: which vendor generation delivers the strongest expected-return signal after transaction costs? **RQ4**: are LLM rankings mechanical rediscoveries of known style sorts, or do they combine market priors with model-specific tilts? Our contributions are designed to make that benchmark auditable: (i) a self-contained evaluation of zero-shot LLM cross-sectional rankings using FF6 spanning tests, within-universe style fingerprints, common-window checks, and transaction costs; (ii) a cross-model benchmark across 11 frontier LLMs, non-LLM factor/ML baselines, and expected-return as the primary objective; (iii) evidence that strong LLMs in this sample deliver large FF6 alphas with a shared growth–momentum tilt but vendor-specific loadings; and (iv) an input ablation showing that market summaries dominate fundamentals in both performance and factor-profile preservation.

2 RELATED WORK

LLM-based equity analysis and portfolio construction. LLMs have been shown to extract financial signals from news and analyst reports [21, 30], financial statements [22], and fundamentals-plus-news data for equity rating [34], though Papasotiriou et al. evaluate only one model and do not backtest investable portfolios. Recent systems move toward full stock selection and portfolio construction: MarketSenseAI and its agentic extension use GPT-4 for holistic selection [10, 11]; TradExpert combines expert LLMs for prediction and ranking [6]; FinMem, TradingAgents, FinCon, and MASS introduce memory, debate, manager–analyst, or scaling-simulation agent designs for trading and portfolio management [16, 47, 49, 50]; prompt-guided GuruAgents encode investor personas into tool-using LLM agents [25]; and Lee et al. [28] integrate LLM views into Black–Litterman and report model-specific “investment styles.” Complementary threads study LLM-enhanced momentum [1], region-specific prediction [44], exploratory portfolio selection [38, 46], and decision-focused optimization layers [19, 23]. We instead study whether frontier LLMs can act as zero-shot *listwise cross-sectional rankers*—outputting a complete ordering rather than ratings or weights—evaluated via backtesting and factor-analytic diagnostics.

Cross-sectional ranking and learning to rank. Framing stock selection as ranking predates LLMs: traditional factor research sorts on human-defined characteristics [8, 20], and modern ML learns ranking signals from characteristic panels [2, 15]. An equity-specific L2R literature covers sentiment-driven ListNet/RankNet [42], Relational Stock Ranking [12], STHAN-SR hypergraph attention [41], pointwise/pairwise/listwise objectives [35, 51], label design [17], and temporal stock representations [18]. In NLP, LLMs

themselves act as zero-shot rankers through RankGPT-style permutation generation and later pairwise, setwise, and listwise prompting [36, 37, 43, 52]. Our contribution is to elicit full listwise cross-sectional rankings *zero-shot* from frontier LLMs without any task-specific financial training.

Factor generation, multi-agent systems, and recommendation. LLMs have also been used to generate predictive factors, mine decay-resistant alphas, and automate quant factor–model R&D [4, 29, 45], coordinate financial decisions via multi-agent systems [3, 31, 48], and query financial databases through constrained function-calling interfaces [24]; see Nie et al. [33] for a survey. Our factor analysis (Section 4.3) treats LLM rankings themselves as implicit factors and tests their spanning by classical risk premia. A related but distinct literature on *personalized* asset recommendation [5, 27, 39, 40] targets user-specific utility; our study instead asks how LLM rankings behave as cross-sectional asset-pricing operators on a fixed investable universe.

3 METHODOLOGY

3.1 Listwise Ranking Formulation

Among pointwise, pairwise, setwise, and listwise L2R paradigms [36, 37, 52], we adopt the *listwise* approach: the LLM receives data for all n stocks simultaneously and returns a single permutation, avoiding the $\binom{n}{2}$ cost of pairwise calls and the coarseness of classification. Rankings preserve the relative information needed for portfolio construction while discarding the magnitude noise that dominates raw return prediction [15], and any classification or regression output can be converted to a ranking but not vice versa.

3.2 Universe and Data

Our universe is the 30 largest NASDAQ-100 constituents by market cap (NVDA, AAPL, MSFT, GOOG, AMZN, META, TSLA, and 23 others)—a fixed, concentrated evaluation panel that is tractable within a single prompt while still supporting cross-sectional ranking. This is a benchmark panel, not a survivorship-free investable universe: fixing the cross-section isolates model-ranking behavior and prompt scalability, while Section 5 treats broader rolling universes as a necessary extension. Each stock-month is described by two JSON blocks: **Fundamentals**—up to 40 S&P Capital IQ items (revenue, net income, EPS, total assets, operating cash flow, etc.) with their four most recent quarterly values; and **Market summary**—nine fields covering 1m/3m returns, 1m/3m annualized volatility, 20d/60d MA gaps, average volume, volume z-score, and drawdown from high. All inputs are point-in-time: only information available before the first day of the prediction month is included.

3.3 Listwise Ranking Protocol

For each month t , prediction mode $m \in \{\text{return, Sharpe, Sortino}\}$, and model M , we (1) build a JSON payload with the fundamentals and market summary for all $n = 30$ stocks as of t , (2) prompt M with a system instruction—“*Rank the following stock tickers by their expected [return / Sharpe / Sortino] over the next rebalancing period. Use ascending order (1 = highest). Return JSON only.*”—and (3) parse the JSON response into a ranking $r_{i,t} \in \{1, \dots, n\}$. All API calls use temperature = 0 for reproducibility, and malformed

JSON responses are discarded rather than repaired by hand. This design makes the benchmark intentionally austere: differences across models arise from the same point-in-time data, the same listwise prompt, and the same deterministic decoding protocol. The user payload is enclosed in `<DATA_JSON>` tags and contains three keys: fundamentals, market, and notes. The market block includes recent return, volatility, moving-average gap, volume, and drawdown summaries; the notes block records the as-of month and quarter ordering.

3.4 Models

We evaluate 11 frontier LLMs from three vendors—OpenAI (GPT-4o/4o mini/4.1/4.1 mini/4.1 nano, o3, o4-mini), Google (Gemini 2.5 Flash/Pro), and Anthropic (Claude Sonnet 4, Opus 4)—spanning multiple generations and a wide cost range. For the asset-pricing narrative we highlight one representative model per vendor family—o3, Gemini 2.5 Pro, and Claude Opus 4—while retaining the full 11-model benchmark in the tables. Every prompt contains only point-in-time data available at the as-of date, reducing input-level lookahead; however, for backtesting we do *not* rely solely on the official model-documentation “knowledge cutoff” date. Instead, the admissible boundary is the latest model release or documented update date available at backtest time whenever that is later than the static knowledge cutoff, and we include only full calendar months after that boundary. Expected return is the primary objective throughout; Sharpe and Sortino prompts are reported as robustness checks on risk-adjusted wording.

3.5 Backtest Design and Metrics

Each monthly ranking is converted into equal-weighted quintile portfolios: **Q1** (stocks ranked 1–6), **Q5** (ranked 25–30), and a dollar-neutral long–short **Q1–Q5**. Portfolios are rebalanced on the third trading day of each month (lag = 2 days) with a one-way transaction cost of 40 bps on turnover. We compare against equal-weighted (**UNIV_EW**) and cap-weighted (**UNIV_CAPW**) benchmarks over the same universe. As non-LLM ranker controls, we also evaluate single-characteristic factor sorts (12–1 momentum, 1-month reversal, 60-day low volatility, earnings yield, ROE, and conservative investment), an equal-weighted composite factor ranker, and expanding-window Ridge/Random-Forest return rankers trained only on prior months. All portfolio metrics—CAGR, annualized volatility, Sharpe, Sortino, maximum drawdown (MDD), Calmar, and hit ratio—are computed from daily returns and annualized where applicable to enable comparability across different backtest lengths. To measure ranking quality directly we compute the Spearman Rank IC between the predicted ranking and the realized return ranking over the same monthly holding period, together with its annualized information ratio $ICIR = IC/\sigma_{IC} \times \sqrt{12}$. Because parsed evaluation coverage differs across models and recent model windows can be short, we do not treat annualized CAGR as a standalone acceptance criterion. Instead, a ranking is credible only when portfolio separation, rank IC, FF6 spanning tests, same-window controls, and within-vendor generation comparisons point in the same direction. Table 1 summarizes the tested model

Table 1: Model coverage, conservative evaluation starts, and approximate monthly experiment costs.

Model	Vendor	Boundary	Start	Cost
GPT-4o	OpenAI	2024-05	2024-06	\$1.03
GPT-4o mini	OpenAI	2024-07	2024-08	\$1.03
GPT-4.1	OpenAI	2025-04	2025-05	\$1.23
GPT-4.1 mini	OpenAI	2025-04	2025-05	\$0.33
GPT-4.1 nano	OpenAI	2025-04	2025-05	\$0.04
o4-mini	OpenAI	2025-04	2025-05	\$0.55
o3	OpenAI	2025-04	2025-05	\$0.93
Gemini 2.5 Flash	Google	2025-06	2025-07	\$0.14
Gemini 2.5 Pro	Google	2025-06	2025-07	\$0.59
Claude Sonnet 4	Anthropic	2025-05	2025-06	\$1.28
Claude Opus 4	Anthropic	2025-05	2025-06	\$6.38

set, conservative boundary used for filtering, first admissible evaluation month, and approximate per-experiment API cost, making the cost–performance comparisons in RQ3 auditable.

4 RESULTS

4.1 Q1 Performance and Quintile Monotonicity

Figure 2 displays the CAGR of each quintile, the central diagnostic for whether the entire ranking produces cross-sectional separation rather than a single lucky top pick. **o3 is the strongest top-quintile ranker**, reaching a Q1 CAGR of 280.4% (Sharpe = 4.41) in the expected-return mode—far above the benchmarks—with a cleanly monotonic Q1→Q5 pattern. Gemini 2.5 Flash, Gemini 2.5 Pro, and GPT-4.1 show similarly strong top-quintile performance, with expected-return Q1 CAGRs above 240%. In contrast, **GPT-4o mini produces negative Q1 returns** and negative long–short spreads, a signal worse than random in this evaluation window. This first screen answers whether the LLM output is an economically ordered ranking at all before applying factor diagnostics.

4.2 Long–Short Spread

Table 2 reports the market-neutral Q1–Q5 long–short portfolio together with rank IC, with boldface marking the best model within each metric–objective column. o3 is the strongest expected-return spread model in the conservative window, reaching a 466.0% net long–short CAGR, Sharpe 5.54, Sortino 10.70, and IC 0.594 after charging both long- and short-leg turnover. Gemini 2.5 Pro (458.8%, Sharpe 4.99, IC 0.577), Gemini 2.5 Flash (380.7%, Sharpe 4.79, IC 0.513), and GPT-4.1 (367.6%, Sharpe 4.72, IC 0.549) also deliver large positive spreads. GPT-4o and GPT-4o mini produce negative long–short returns in the conservative window, confirming that model choice can flip the sign of the induced cross-sectional prior. Because these are annualized short-window portfolio statistics, the relevant evidence is their agreement with IC, ensemble, same-window, and factor-spanning diagnostics, not the CAGR magnitude alone.

An offline average-rank ensemble check shows that the signal is not purely a single-model artifact. Table 3 averages expected-return ranks across model sets on their common admissible months. The nine-model consensus across GPT-4.1, o-series, Gemini 2.5, and Claude 4 families reaches a 474.2% net long–short CAGR, Sharpe 5.44, and IC 0.549 from 2025-07 to 2025-12, while the OpenAI reasoning consensus reaches Sharpe 5.39.

Quintile Monotonicity Check

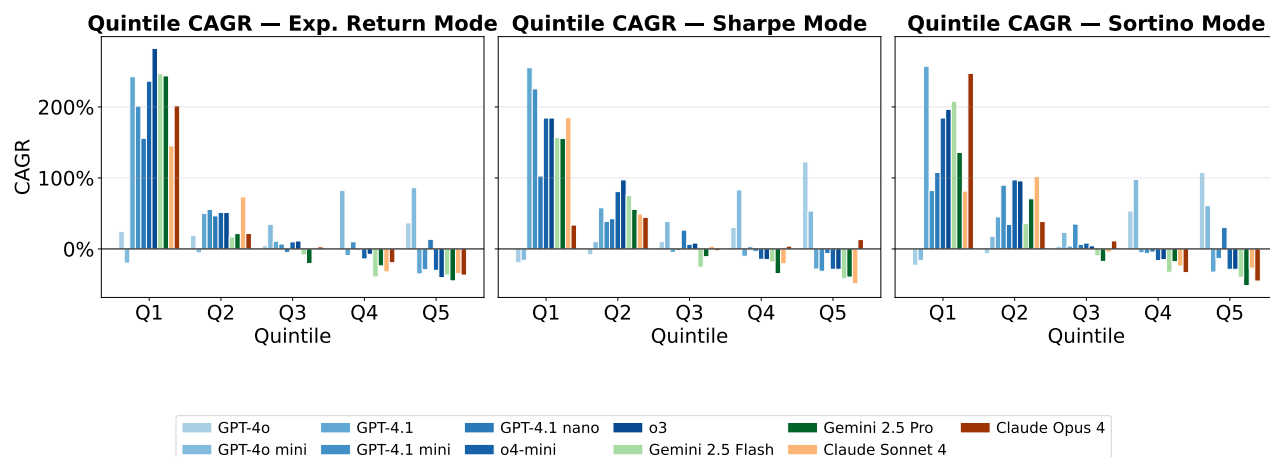


Figure 2: Quintile CAGR by model and prompt objective in each model’s conservative evaluation window.

Table 2: Conservative-window Q1–Q5 long–short performance and rank IC, net of 40 bps one-way costs on both legs. Bold indicates the best value in each column.

Ranker	Exp. Return					Sharpe					Sortino				
	CAGR	Shp	Sort	MDD	IC	CAGR	Shp	Sort	MDD	IC	CAGR	Shp	Sort	MDD	IC
GPT-4o	−22.7	−0.62	−0.83	−57.3	0.090	−70.3	−3.31	−3.89	−86.0	−0.264	−69.8	−3.31	−3.92	−85.7	−0.281
GPT-4o mini	−63.1	−4.02	−4.70	−77.0	−0.303	−52.7	−2.92	−3.63	−66.1	−0.241	−55.7	−3.15	−3.86	−69.6	−0.255
GPT-4.1	367.6	4.72	8.37	−11.8	0.549	342.7	4.85	9.29	−11.3	0.540	372.1	4.90	9.30	−10.8	0.510
GPT-4.1 mini	260.3	4.09	7.64	−15.9	0.449	314.0	4.82	8.96	−8.5	0.498	81.2	2.41	4.06	−15.7	0.306
GPT-4.1 nano	100.4	2.82	4.63	−8.8	0.294	88.2	2.75	4.56	−11.6	0.300	42.1	1.80	2.71	−8.9	0.230
o4-mini	333.0	4.65	8.39	−11.4	0.529	258.7	4.48	7.75	−8.5	0.524	258.7	4.48	7.75	−8.5	0.526
o3	466.0	5.54	10.70	−8.7	0.594	258.7	4.48	7.75	−8.5	0.526	274.1	4.61	8.07	−8.5	0.561
Gem. 2.5 Flash	380.7	4.79	8.81	−14.1	0.513	298.5	4.64	8.40	−7.7	0.509	356.7	5.22	10.65	−7.1	0.482
Gem. 2.5 Pro	458.8	4.99	9.15	−11.4	0.577	270.2	4.55	8.37	−6.4	0.511	329.1	5.86	12.38	−4.9	0.569
Cl. Sonnet 4	231.9	3.99	7.24	−9.9	0.534	389.3	5.78	11.80	−6.0	0.588	113.5	2.91	4.77	−14.8	0.539
Cl. Opus 4	312.9	4.56	8.20	−13.0	0.483	−1.6	0.06	0.09	−33.7	0.283	444.9	5.74	11.63	−6.0	0.749

Table 3: Conservative-window average-rank ensemble robustness. Consensus ranks are evaluated as the same net 40 bps Q1–Q5 portfolio.

Consensus	Models	Start	Mo.	LS CAGR	LS Shp	IC
Nine-model consensus	9	2025-07	6	474.2	5.44	0.549
OpenAI 4.1 consensus	3	2025-05	8	306.8	4.49	0.486
OpenAI reasoning consensus	2	2025-05	8	428.9	5.39	0.567
Gemini 2.5 consensus	2	2025-07	6	454.2	4.86	0.542
Claude 4 consensus	2	2025-06	7	317.2	4.57	0.560
Low-cost consensus	3	2025-07	6	379.4	4.89	0.515

4.3 Factor Analysis of LLM Rankings

The central asset-pricing question is whether LLM-induced orderings behave like factor-like cross-sectional priors. If rankings simply re-discover known risk premia, their long–short returns should be spanned by standard factor models; if not, they should exhibit distinct loadings and alphas. We address this with two analyses: (i) time-series *spanning regressions* of Q1–Q5 daily returns on established factors [7–9], and (ii) *characteristic-based style analysis* mirroring the classical factor workflow (sort, form long–short,

test alpha) but replacing the human sort with the LLM’s ranking. Section 4.4 applies the same logic to fundamentals-only and market-only input ablations.

Spanning test. For each expected-return LLM ranker, we regress gross daily Q1–Q5 returns on FF6 factors [13] with Newey–West HAC SE (5 lags); Table 4 reports annualized alpha, factor loadings, and adjusted R^2 . Three findings stand out: (i) *winning LLMs have large, highly significant alphas*—o3 $\alpha = 351\%$ ($t = 5.7$), Gemini 2.5 Pro $\alpha = 340\%$ ($t = 4.7$), and Gemini 2.5 Flash $\alpha = 285\%$ ($t = 4.7$)—so their returns are not mechanically spanned by the standard factor zoo in this benchmark; (ii) adjusted R^2 ranges from 0.08 to 0.52, indicating that standard factors explain part, but not all, of the spread-return variation; (iii) *failing models have significantly negative alphas* (GPT-4o mini $\alpha = -55\%$, $t = -4.5$), i.e., structurally counter-factorial during the AI-rally period.

Style fingerprint. Because global Fama–French factors are built on the full US cross-section while our universe is 30 mega-cap NASDAQ stocks, they may understate within-universe style tilts. We

Table 4: FF6 spanning tests for conservative-window Q1–Q5 returns. α is annualized percent; stars use Newey–West HAC SE: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Model	α (%)	$t(\alpha)$	β_{MKT}	β_{SMB}	β_{HML}	β_{RMW}	β_{CMA}	β_{MOM}	R^2_{adj}
GPT-4o	−14.2	−0.54	0.34***	0.07	−0.28	−0.03	−0.50*	0.17	0.08
GPT-4o mini	−54.8***	−4.52	−0.06	−0.14	0.21	0.07	0.58***	−0.38***	0.13
GPT-4.1	250.2***	4.73	1.23***	−0.48*	−0.36	−0.92***	−0.23	0.36	0.45
GPT-4.1 mini	225.5***	3.38	0.55**	0.12	0.10	−0.61*	−0.61	0.69**	0.28
GPT-4.1 nano	96.6**	2.39	0.74***	−0.18	−0.11	0.22	−0.39	−0.04	0.10
o4-mini	230.1***	4.61	1.24***	−0.45*	−0.31	−0.69***	−0.42	0.32	0.44
o3	350.8***	5.71	1.15***	−0.47*	−0.29	−0.67***	−0.22	0.25	0.35
Gem. 2.5 Flash	284.7***	4.67	1.33***	−0.32	−0.08	−0.32	−0.54	0.58*	0.48
Gem. 2.5 Pro	340.2***	4.71	1.49***	−0.47	−0.27	−0.48*	−0.40	0.51	0.50
Cl. Sonnet 4	182.9***	5.02	0.95***	−0.46*	−0.09	−0.40	−1.02***	0.45**	0.52
Cl. Opus 4	261.0***	3.33	1.04***	0.01	0.22	0.26	−1.39**	0.12	0.25

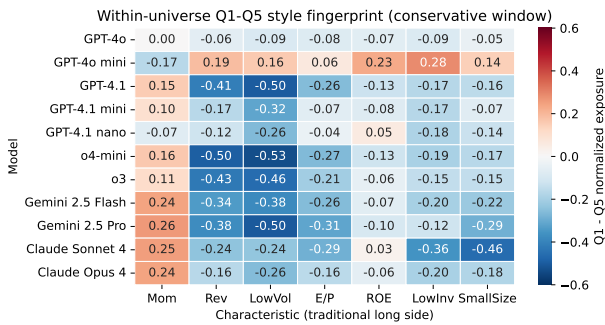
therefore rank stocks each month on each characteristic (normalized to $[0, 1]$ so that +1 is the factor’s traditional “long” side) and average the rank within each LLM’s Q1 and Q5. Figure 3 visualizes the Q1–Q5 style fingerprint for the expected-return mode.

A consistent pattern emerges across the strongest models: large positive momentum tilts (+0.23 to +0.32) combined with *negative* exposures to low-volatility, value, profitability, and conservative-investment—a “growth–momentum chaser” style that worked in the 2024–2025 AI rally and matches their FF6 loadings in Table 4. GPT-4o mini displays the mirror image (negative momentum, positive reversal and volatility), explaining its anti-monotonic returns: it systematically picks defensive, recently-weak stocks in a period when the opposite was rewarded. Within-universe factor-mimicking correlations show the same qualitative pattern at the daily-return level.

Together these results imply that (a) LLM long–short portfolios in this panel are not fully recovered by linear combinations of classical style factors—consistent with LLM-based factor generation [4]—and (b) each LLM embeds a distinct implicit prior over which characteristics predict returns. Selecting an LLM therefore changes the implicit factor exposure being evaluated.

4.4 Input Ablation and Factor Loadings

The factor-prior interpretation also asks which information channel generates the implicit ranking profile. We re-run every model–objective pair with fundamentals only, market summaries only, or

**Figure 3: Within-universe Q1–Q5 style fingerprint in the expected-return mode.** Positive cells tilt toward the characteristic’s traditional long side.**Table 5: Input ablation diagnostics under conservative model windows.** CAGR and IC are reported in percent; Panel C stars use Newey–West HAC SE: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Panel A: Expected-return performance					
Input	Mean LS CAGR	Mean LS Shp	Mean IC	N	
Fund. + Market	256.9	3.23	39.2	11	
Market only	313.8	3.38	48.7	9	
Fund. only	47.6	1.32	11.4	9	
Panel B: Paired tests					
Comparison	Metric	Both	Ablated	Δ	p
Both - Market	LS CAGR	218.2	294.3	−76.1	0.154
Both - Market	LS Shp	2.68	3.21	−0.53	0.478
Both - Market	IC	33.4	43.5	−10.1	0.154
Both - Fund.	LS CAGR	218.2	20.4	197.8	0.021
Both - Fund.	LS Shp	2.68	0.53	2.15	0.096
Both - Fund.	IC	34.1	7.4	26.7	0.053
Panel C: Representative FF6 loadings					
Model	Input	α	β_{HML}	β_{CMA}	β_{MOM}
o3	Fund. + Market	350.8	−0.29	−0.22	0.25
o3	Market only	317.4	−0.19	−0.28	0.46**
o3	Fund. only	84.6	−0.50**	−0.40	0.31
Gem. 2.5 Pro	Fund. + Market	340.2	−0.27	−0.40	0.51
Gem. 2.5 Pro	Market only	414.2	−0.62	0.50	0.47
Gem. 2.5 Pro	Fund. only	76.2	−0.27	−0.69	0.38
Cl. Opus 4	Fund. + Market	261.0	0.22	−1.39**	0.12
Cl. Opus 4	Market only	555.9	−0.20	−0.45	0.13
Cl. Opus 4	Fund. only	−7.2	−0.21	−1.26***	0.18

both inputs, then repeat the FF6 spanning test for one representative model per vendor in expected-return mode. Table 5 shows that market-only prompts retain most of the measured signal and slightly exceed the full prompt on average in the conservative window, whereas fundamentals-only prompts are much weaker; Panel C links that performance gap to changes in HML, CMA, and momentum loadings for representative models. This supports the interpretation that recent market summaries, rather than sparse accounting panels alone, carry most of the observed ranking signal in this narrow universe.

4.5 Rank IC Analysis

Because Table 2 reports rank IC next to portfolio outcomes, it also checks whether returns arise from genuine cross-sectional ordering rather than a few lucky names. Across the LLM benchmark, **o3** achieves the highest expected-return IC (0.594), followed by Gemini 2.5 Pro (0.577), GPT-4.1 (0.549), and Claude Sonnet 4 (0.534). GPT-4.1 mini remains positive after patching malformed months

Long–Short (Q1–Q5) Cumulative Wealth

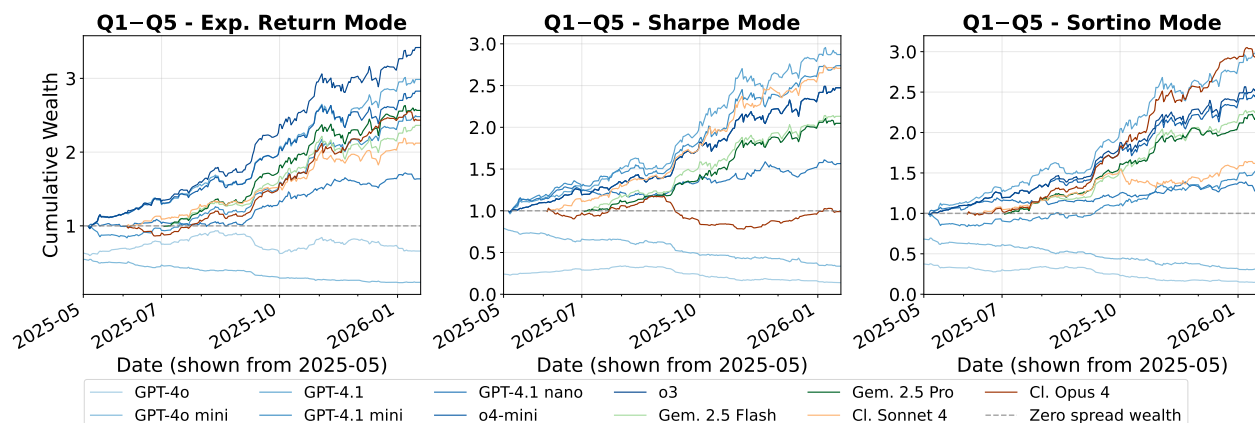


Figure 4: Cumulative Q1–Q5 wealth by model and prompt objective, shown from 2025-05 after compounding each path from its conservative start; the dashed line marks zero spread wealth.

(IC = 0.449), while GPT-4o mini is negative across all modes—its rankings are systematically inversely correlated with realized outcomes.

4.6 Prediction Mode and Cumulative Returns

Across the included models, the **expected-return mode remains the clearest primary benchmark**: it directly targets the holding-period objective and produces the strongest spread for several of the top rankers, while Sharpe/Sortino prompts act more like style-control variants. Sharpe/Sortino prompts win for specific models (e.g., GPT-4.1 reaches its best Q1 CAGR of 257.3% in Sortino mode) but produce more variable results overall, suggesting expected-return prompts remain the clearest primary benchmark. Figure 4 shows the same pattern in cumulative wealth: the stronger post-2025 rankers, including o3, Gemini 2.5, GPT-4.1-family, and Claude 4 models, accumulate positive spreads, while GPT-4o-family models lose money.

4.7 Protocol Checks

Several design choices make the benchmark conservative. First, all LLM and non-LLM rankers are evaluated on the same monthly universe, holding-period definition, 2-day execution lag, and 40 bps one-way transaction cost charged to both long- and short-leg turnover. The performance rows in Table 2 therefore share the same market data, universe, rebalance schedule, and transaction-cost accounting, so differences reflect the induced rankings rather than different portfolio mechanics. Second, the model table fixes the information boundary induced by knowledge cutoffs. Because newer models have shorter admissible post-cutoff windows, we avoid interpreting raw CAGR alone and instead emphasize within-vendor generation comparisons, rank IC, same-window non-LLM controls, and FF6 alphas. The conservative boundary prevents newer preview models from borrowing pre-release 2025 months and makes the negative GPT-4o-family results directly visible. After patching malformed non-deprecated OpenAI outputs, all included model-mode panels have at least 20 ranked names in each evaluated month.

Third, the listwise protocol uses one deterministic call per model-month-objective rather than prompt search, voting, or hand repair. Malformed JSON is discarded rather than manually corrected, and temperature is fixed at zero. This keeps the reported differences attributable to the ranker induced by a frontier model under a common prompt, not to downstream engineering. It also makes the benchmark a useful lower bound for future agentic systems: an agentic ranker, pairwise tournament, or ensemble should be compared against these single-call listwise results under the same information set and cost accounting.

5 DISCUSSION

Scope of the evidence. The results should be read as a stress test of model-induced rankings under a fixed information set. The fixed 30-stock NASDAQ panel, short post-release windows, simplified shorting assumptions, and possible hidden memorization channels limit direct claims about deployable trading performance. Their value is instead diagnostic: the same protocol reveals which LLMs induce monotone rankings, positive IC, distinct style loadings, and residual alphas under conservative timing.

Model generation effect. The most striking finding is the gap between generations and vendors: within OpenAI, o3’s expected-return IC of 0.594 and GPT-4.1’s 0.549 greatly exceed GPT-4o’s 0.090 and GPT-4o mini’s -0.303 ; within Google, Gemini 2.5 Pro and Flash both deliver strong positive ICs despite short conservative windows. The pattern is directionally consistent with model capability: newer reasoning models and larger frontier models tend to produce stronger rankings, although price is not a sufficient statistic because smaller models such as Gemini 2.5 Flash and GPT-4.1 mini remain competitive.

Ranking vs. classification. Full rankings expose signal structures that classification would hide: GPT-4o mini’s *anti-monotonic* quintile returns would be invisible under a 5-category Buy/Hold/Sell scheme, and o3’s market-neutral long–short Sharpe of 5.54 is only attainable when rankings separate both the top and bottom of the cross-section. In our setting, the bottom quintile is as informative as the top: strong models avoid the same stocks that weak

models over-rank, converting a long-only selection signal into an investable spread. Buy/Hold/Sell labels or top- k hit rates would therefore understate both positive and negative model skill.

Cost-effectiveness and reproducibility. Gemini 2.5 Flash, GPT-4.1 mini, and GPT-4.1 nano offer strong cost-performance trade-offs among included models: all are inexpensive relative to larger frontier or reasoning models yet deliver positive spreads after conservative-window filtering. High-cost models such as Claude Opus 4 are less attractive on a spread-return basis. The relevant unit is not leaderboard capability but ranking quality per dollar under a fixed information set.

Cost-Effectiveness Analysis

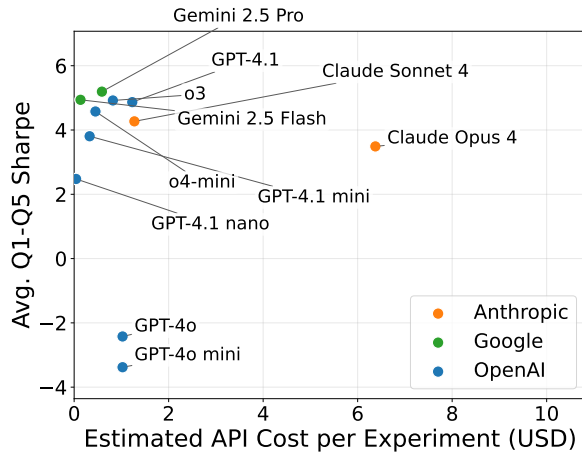


Figure 5: Cost versus average Q1–Q5 Sharpe for tested LLM rankers.

Cost should be reported at the experiment level rather than only per token because pairwise or agentic variants can multiply API usage quickly.

Risk-adjusted prompts as implicit risk control. Prompting for Sharpe/Sortino acts as implicit risk control even without weights or an optimizer: the risk-adjusted objectives often reduce drawdowns or change which names enter Q1 and Q5. The factor analysis (Section 4.3) confirms a genuine style shift, moving the implicit style away from a pure “momentum $\times$ growth” profile toward more risk-aware rankings for some models. However, the risk prompts do not uniformly improve ranking quality. They often reduce drawdowns by dampening the high-momentum growth tilt, but they can also dilute or redirect the strongest return signal, as seen in the model-specific differences between expected-return, Sharpe, and Sortino prompts. Prompt choice should therefore be viewed as a style-control knob rather than a free improvement in objective alignment.

IC as diagnostic, vendor risk profiles, and limitations. Mean IC and long-short Sharpe are near-linearly related ($r = 0.90$, $p < 0.001$), making IC a reliable ex-ante diagnostic. Anthropic long-short portfolios show relatively shallow drawdowns in conservative windows, while the GPT-4o family shows much deeper losses and negative ICs. Remaining limitations are important for follow-on work: post-cutoff windows reduce direct leakage from evaluated holding-period returns, but broader LLM memorization, hidden

lookahead channels [14, 32], and market reflexivity in historical AI backtests [26] warrant formal testing; backtest windows differ outside the common-window robustness check; and LLM outputs can drift across API versions despite temperature = 0. Some historical model IDs are preview, date-pinned, or currently unavailable for live reruns, so we distinguish cached benchmark evidence from future live-API reproducibility. We therefore interpret the results as a controlled model-comparison benchmark rather than evidence of production tradability. A natural benchmark extension is to add pointwise LLM scoring, pairwise or setwise tournaments, ensembles, and lightweight agentic rankers on the identical information set. The fixed listwise protocol isolates a minimum viable cross-sectional ranker before adding retrieval, debate, tool use, or portfolio optimization layers. Negative results are equally informative: GPT-4o mini persistently ranks the cross-section in the wrong direction, producing negative IC and anti-monotonic quintiles rather than merely noisy alpha. For benchmark design, rank IC, portfolio spreads, and style diagnostics should therefore be reported together: IC tests ordering quality, spreads test economic investability, and factor diagnostics test whether apparent alpha is just familiar style compensation.

6 CONCLUSION

We studied LLMs as implicit factor-like priors for cross-sectional stock ranking, bridging LLM capability evaluation and factor diagnostics. The central lesson is that model choice is an asset-pricing choice: switching the LLM changes not only ranking accuracy, but also the growth, momentum, value, profitability, and investment exposures being tested. Across 11 included models, expected-return prompting is the primary lens: o3 reaches a 466.0% net long-short CAGR, Sharpe 5.54, and IC 0.594 after charging both long- and short-leg turnover, while GPT-4o and GPT-4o mini produce weak or negative signals. The input ablation shows that market summaries carry most of the pricing signal and factor-profile coherence, whereas fundamentals-only prompts weaken both alpha and style structure. For finance benchmarks, the implication is to report ordering quality, investable spread performance, and factor diagnostics together rather than treating an LLM ranking as a black-box forecast. Future work should extend to broader rolling universes, live model refreshes, and disclosure-text information axes while preserving the same audit trail from prompt to ranking, portfolio, and factor exposure.

REFERENCES

- [1] Anic, N., Barbon, A., Seiz, R., and Zarattini, C. (2025). ChatGPT in systematic investing: Enhancing risk-adjusted returns with LLMs. *arXiv preprint arXiv:2510.26228*.
- [2] Bryzgalova, S., Pelger, M., and Zhu, J. (2025). Forest through the trees: Building cross-sections of stock returns. *The Journal of Finance*, 80(5):2447–2506.
- [3] Cheng, J. and Chin, P. (2024). AAPM: Large language model agent-based asset pricing models. *arXiv preprint arXiv:2409.17266*.
- [4] Cheng, Y., Zeng, Y., and Zou, J. (2024). Harnessing ChatGPT for predictive financial factor generation: A new frontier in financial analysis and forecasting. *The British Accounting Review*, 101507.
- [5] Chung, M., Lee, J., Lee, Y., and Kim, W. C. (2025). Mean variance efficient collaborative filtering for stock recommendations. *Proceedings of the 6th ACM International Conference on AI in Finance (ICAIF '25)*, pages 806–813. arXiv:2306.06590.
- [6] Ding, Q., Shi, H., Guo, J., and Liu, B. (2025). TradExpert: Revolutionizing trading with mixture of expert LLMs. *ICLR 2025 Workshop on Advances in Financial AI*. arXiv:2411.00782.

- [7] Carhart, M. M. (1997). On persistence in mutual fund performance. *The Journal of Finance*, 52(1):57–82.
- [8] Fama, E. F. and French, K. R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1):3–56.
- [9] Fama, E. F. and French, K. R. (2015). A five-factor asset pricing model. *Journal of Financial Economics*, 116(1):1–22.
- [10] Fatouros, G., Metaxas, K., Soldatos, J., and Kyriazis, D. (2024). Can large language models beat wall street? Evaluating GPT-4’s impact on financial decision-making with MarketSenseAI. *Neural Computing and Applications*, pages 1–26. doi:10.1007/s00521-024-10613-4.
- [11] Fatouros, G., Metaxas, K., Soldatos, J., and Karathanassis, M. (2025). MarketSenseAI 2.0: Enhancing stock analysis through LLM agents. *IEEE International Conference on Data Mining Workshops (ICDMW ’25)*, pages 883–892. arXiv:2502.00415.
- [12] Feng, F., He, X., Wang, X., Luo, C., Liu, Y., and Chua, T.-S. (2019). Temporal relational ranking for stock prediction. *ACM Transactions on Information Systems*, 37(2):Article 27.
- [13] French, K. R. (2025). Data library: Daily research returns, 5 factors, and momentum. https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html.
- [14] Gao, Z., Jiang, W., and Yan, Y. (2025). A test of lookahead bias in LLM forecasts. *arXiv preprint arXiv:2512.23847*.
- [15] Gu, S., Kelly, B., and Xiu, D. (2020). Empirical asset pricing via machine learning. *The Review of Financial Studies*, 33(5):2223–2273.
- [16] Guo, T., Shen, H., Huang, J., Mao, Z., Luo, J., Chen, B., Chen, Z., Liu, X., Xia, B., Liu, L., Ma, Y., and Zhang, M. (2025). MASS: Multi-agent simulation scaling for portfolio construction. *arXiv preprint arXiv:2505.10278*.
- [17] Hwang, Y., Park, J., Lee, Y., and Lim, D.-Y. (2023). Stop-loss adjusted labels for machine learning-based trading of risky assets. *Finance Research Letters*, 58(A):104285.
- [18] Hwang, Y., Zohren, S., and Lee, Y. (2024). Temporal representation learning for stock similarities and its applications in investment management. *arXiv preprint arXiv:2407.13751*.
- [19] Hwang, Y., Kong, Y., Zohren, S., and Lee, Y. (2025). Decision-informed neural networks with large language model integration for portfolio optimization. *arXiv preprint arXiv:2502.00828*.
- [20] Jegadeesh, N. and Titman, S. (1993). Returns to buying winners and selling losers: Implications for stock market efficiency. *The Journal of Finance*, 48(1):65–91.
- [21] Kim, S., Kim, S., Kim, Y., Park, J., Kim, S., Kim, M., Sung, C. H., Hong, J., and Lee, Y. (2023). LLMs analyzing the analysts: Do BERT and GPT extract more value from financial analyst reports? *Proceedings of the Fourth ACM International Conference on AI in Finance (ICAIF ’23)*, pages 383–391.
- [22] Kim, A., Muhn, M., and Nikolaev, V. (2024). Financial statement analysis with large language models. *Working Paper*, University of Chicago Booth School of Business.
- [23] Kim, J., Choi, S., Lee, Y., Kim, Y., Choi, Y., and Lee, Y. (2025a). Decision by supervised learning with deep ensembles: A practical framework for robust portfolio optimization. *arXiv preprint arXiv:2503.13544*.
- [24] Kim, J., Kim, Y., Lee, Y., and Byun, H. (2025b). FinAI Data Assistant: LLM-based financial database query processing with the OpenAI function calling API. *arXiv preprint arXiv:2510.14162*.
- [25] Kim, Y., Lee, Y., Kim, J., and Lee, Y. (2025c). GuruAgents: Emulating wise investors with prompt-guided LLM agents. *arXiv preprint arXiv:2510.01664*.
- [26] Koijen, R. S. J. and Levy, B. (2026). Assessing the benefits of optimized agentic AI systems for asset pricing. *SSRN working paper*.
- [27] Lee, Y., Kim, Y., Sanz-Cruzado, J., McCreadie, R., and Lee, Y. (2024). Stock recommendations for individual investors: A temporal graph network approach with mean-variance efficient sampling. *Proceedings of the 5th ACM International Conference on AI in Finance (ICAIF ’24)*, pages 795–803. doi:10.1145/3677052.3698662.
- [28] Lee, Y., Kim, Y., Kim, J., Kim, S., and Lee, Y. (2025). LLM-enhanced Black-Litterman portfolio optimization. *arXiv preprint arXiv:2504.14345*.
- [29] Li, Y., Yang, X., Yang, X., Xu, M., Wang, X., Liu, W., and Bian, J. (2025). R&D-Agent-Quant: A multi-agent framework for data-centric factors and model joint optimization. *arXiv preprint arXiv:2505.15155*.
- [30] Lopez-Lira, A. and Tang, Y. (2023). Can ChatGPT forecast stock price movements? Return predictability and large language models. *Working Paper*, University of Florida.
- [31] Zhao, T., Lyu, J., Jones, S., Garber, H., Pasquali, S., and Mehta, D. (2025). AlphaAgents: Large language model based multi-agents for equity portfolio constructions. *arXiv preprint arXiv:2508.11152*.
- [32] Merchant, H. and Levy, B. (2025). A fast and effective solution to the problem of look-ahead bias in LLMs. *NeurIPS 2025 Workshop: Generative AI in Finance*. arXiv:2512.06607.
- [33] Nie, Y., Kong, Y., Dong, X., Mulvey, J. M., Poor, H. V., Wen, Q., and Zohren, S. (2024). A survey of large language models for financial applications: Progress, prospects and challenges. *arXiv preprint arXiv:2406.11903*.
- [34] Papasotiriou, K., Sood, S., Reynolds, S., and Balch, T. (2024). AI in investment analysis: LLMs for equity stock ratings. *Proceedings of the 5th ACM International Conference on AI in Finance (ICAIF ’24)*, pages 419–427. arXiv:2411.00856.
- [35] Poh, D., Lim, B., Zohren, S., and Roberts, S. (2021). Building cross-sectional systematic strategies by learning to rank. *The Journal of Financial Data Science*, 3(2):70–86.
- [36] Qin, Z., Jagerman, R., Hui, K., Zhuang, H., Wu, J., Yan, L., Shen, J., Liu, T., Liu, J., Metzler, D., Wang, X., and Bendersky, M. (2024). Large language models are effective text rankers with pairwise ranking prompting. *Findings of NAACL 2024*, pages 1504–1518.
- [37] Ren, R., Wang, Y., Zhou, K., Zhao, W. X., Wang, W., Liu, J., Wen, J.-R., and Chua, T.-S. (2025). Self-calibrated listwise reranking with large language models. *Proceedings of The Web Conference 2025 (WWW ’25)*.
- [38] Romanko, O., Narayan, A., and Kwon, R. H. (2023). ChatGPT-based investment portfolio selection. *SN Operations Research Forum*, 4(4):1–27. doi:10.1007/s43069-023-00277-6.
- [39] Sakurai, K., Ogawa, T., Haseyama, M., Anan, A., and Nakagawa, K. (2026). Risk-aware utility re-ranking for financial asset recommendation. *Proceedings of the 19th ACM International Conference on Web Search and Data Mining (WSDM ’26)*.
- [40] Sanz-Cruzado, J., Droukas, N., and McCreadie, R. (2024). FAR-Trans: An investment dataset for financial asset recommendation. *arXiv preprint arXiv:2407.08692*.
- [41] Sawhney, R., Agarwal, S., Wadhwa, A., Derr, T., and Shah, R. R. (2021). Stock selection via spatiotemporal hypergraph attention network: A learning to rank approach. *Proceedings of the AAAI Conference on Artificial Intelligence*, 35(1):497–504.
- [42] Song, Q., Liu, A., and Yang, S. Y. (2017). Stock portfolio selection using learning-to-rank algorithms with news sentiment. *Neurocomputing*, 264:20–28.
- [43] Sun, W., Yan, L., Ma, X., Wang, S., Ren, P., Chen, Z., Yin, D., and Ren, Z. (2023). Is ChatGPT good at search? Investigating large language models as re-ranking agents. *Proceedings of EMNLP 2023*. arXiv:2304.09542.
- [44] Tan, L., Wu, H., and Zhang, X. (2024). Large language models and return prediction in China. *SSRN Working Paper No. 4712248*.
- [45] Tang, Z., Chen, Z., Yang, J., Mai, J., Zheng, Y., Wang, K., Chen, J., and Lin, L. (2025). AlphaAgent: LLM-driven alpha mining with regularized exploration to counteract alpha decay. *Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD ’25)*. arXiv:2502.16789.
- [46] Xie, Q., Han, W., Lai, Y., Peng, M., and Huang, J. (2023). The Wall Street Neophyte: A zero-shot analysis of ChatGPT over multimodal stock movement prediction challenges. *arXiv preprint arXiv:2304.05351*.
- [47] Xiao, Y., Sun, E., Luo, D., and Wang, W. (2024). TradingAgents: Multi-agents LLM financial trading framework. *arXiv preprint arXiv:2412.20138*.
- [48] Yang, H. et al. (2024). FinRobot: An open-source AI agent platform for financial applications using large language models. *arXiv preprint arXiv:2405.14767*.
- [49] Yu, Y. et al. (2024). FinCon: A synthesized LLM multi-agent system with conceptual verbal reinforcement for enhanced financial decision making. *arXiv preprint arXiv:2407.06567*.
- [50] Yu, Y., Li, H., Chen, Z., Jiang, Y., Li, Y., Suchow, J. W., Zhang, D., and Khashanah, K. (2025). FINMEM: A performance-enhanced LLM trading agent with layered memory and character design. *IEEE Transactions on Big Data*. doi:10.1109/TBDATA.2025.3593370.
- [51] Zhang, X., Wu, L., and Chen, Z. (2022). Constructing long-short stock portfolio with a new listwise learn-to-rank algorithm. *Quantitative Finance*, 22(2):321–331.
- [52] Zhuang, S., Zhuang, H., Koopman, B., and Zuccon, G. (2024). A setwise approach for effective and highly efficient zero-shot ranking with large language models. *Proceedings of SIGIR ’24*.

APPENDIX

A PROMPT TEMPLATE

For each prediction month t and mode m , we construct a JSON payload containing fundamental and market data, then send the following instruction to the LLM:

```
Rank the following stock tickers by their expected {mode} over the next rebalancing period. Base your judgment
on reasonable assumptions and available general market knowledge. Use ascending order (1 = highest expected
performance).
Output exactly one ticker per line in the following format:
1: {Ticker1}
2: {Ticker2}
3: {Ticker3}
...
Return JSON only: [{"ticker": "...", "rank": 1}, ...]
```

where {mode} is replaced by “expected return”, “Sharpe Ratio”, or “Sortino Ratio” depending on the prediction objective. The user message then contains a compact JSON payload enclosed in <DATA_JSON> tags. Its top-level keys are: (i) fundamentals—one record per stock, containing up to 40 accounting items selected by the absolute magnitude of the most recent quarter’s value, with four quarterly observations Q1–Q4 (Q1 = oldest, Q4 = most recent); (ii) market—one record per stock with nine recent price/volume summary fields; and (iii) notes—metadata indicating quarter ordering and the as-of month. The market fields are ret_1m, ret_3m, vol_1m, vol_3m, ma20_gap, ma60_gap, avg_volume, volume_z, and drawdown_from_high.

A representative payload fragment is shown below:

```
<DATA_JSON>
{
  "fundamentals": [
    {
      "MONTH": "2025-01",
      "TICKER": "NVDA",
      "items": [
        {"name": "Revenue", "Q1": 26044, "Q2": 30040, "Q3": 35100, "Q4": 39331},
        {"name": "Net Income", "Q1": 12358, "Q2": 16599, "Q3": 19309, "Q4": 22091},
        {"name": "EPS", "Q1": 0.49, "Q2": 0.67, "Q3": 0.78, "Q4": 0.89}
      ]
    }
  ],
  "market": [
    {
      "TICKER": "NVDA",
      "asof": "2025-01-31",
      "ret_1m": 0.1241,
      "ret_3m": 0.3275,
      "vol_1m": 0.4123,
      "vol_3m": 0.3886,
      "ma20_gap": 0.0615,
      "ma60_gap": 0.1432,
      "avg_volume": 42351872,
      "volume_z": 1.2841,
      "drawdown_from_high": -0.0378
    }
  ],
  "notes": {
    "q_order": "Q1 oldest, Q4 most recent",
    "asof_month": "2025-01"
  }
}
</DATA_JSON>
```

In the actual experiments this structure is repeated for all 30 stocks in the universe; only the item names and values differ by stock and month. All API calls use temperature = 0.

B QUINTILE PORTFOLIO PERFORMANCE

Table 6 reports the gross CAGR of each quintile portfolio before transaction costs. Monotonically decreasing values from Q1 to Q5 indicate effective ranking; anti-monotonic patterns ($Q5 > Q1$) indicate counter-productive signals.

Table 6: Gross full-window quintile CAGR (%) by model and prediction mode, before transaction costs. Bold marks the best value in each column within each panel.

Mode	Model	Q1	Q2	Q3	Q4	Q5	Q1–Q5
<i>Panel A: Exp. Return</i>							
	GPT-4o	41.1	28.9	22.5	26.5	49.1	−9.3
	GPT-4o mini	−5.3	28.9	56.1	61.9	45.5	−37.0
	GPT-4.1	92.5	29.4	10.7	0.6	−7.1	105.1
	GPT-4.1 mini	124.7	29.5	5.0	−2.5	−11.5	138.4
	GPT-4.1 nano	77.9	13.6	12.8	−5.8	28.8	34.6
	o4-mini	77.9	37.7	11.3	−10.8	5.5	64.9
	o3	87.0	66.9	5.7	0.5	−26.0	153.2
	Gem. 2.5 Flash	83.0	51.4	26.3	−11.5	−19.5	120.7
	Gem. 2.5 Pro	62.0	38.7	34.0	7.4	−23.2	107.9
	Cl. Sonnet 4	145.0	14.7	8.8	−15.5	−7.0	158.2
	Cl. Opus 4	106.7	33.2	22.2	−16.0	−20.6	156.3
<i>Panel B: Sharpe</i>							
	GPT-4o	−6.2	14.6	12.1	41.3	90.7	−53.5
	GPT-4o mini	−7.0	41.1	51.2	63.3	44.5	−37.4
	GPT-4.1	124.5	35.9	−5.0	−14.7	−1.9	114.9
	GPT-4.1 mini	137.5	26.6	−9.9	13.3	−12.8	168.3
	GPT-4.1 nano	50.1	18.3	20.9	−0.0	18.8	20.5
	o4-mini	73.5	36.0	9.8	−12.7	10.2	40.8
	o3	102.5	51.0	−2.5	−17.5	−3.5	96.6
	Gem. 2.5 Flash	39.8	60.3	−2.2	23.9	15.9	8.3
	Gem. 2.5 Pro	85.4	19.0	21.6	−11.1	27.7	27.0
	Cl. Sonnet 4	104.0	27.2	23.2	9.2	−24.4	147.3
	Cl. Opus 4	66.4	5.9	19.2	18.3	−0.5	53.5
<i>Panel C: Sortino</i>							
	GPT-4o	2.0	21.5	26.4	59.4	77.9	−45.4
	GPT-4o mini	−9.2	37.0	46.0	79.8	56.6	−43.8
	GPT-4.1	93.9	44.7	−5.1	−15.2	22.1	49.7
	GPT-4.1 mini	60.8	60.6	−1.3	24.6	−22.1	95.3
	GPT-4.1 nano	52.7	12.4	10.6	1.5	46.8	−1.2
	o4-mini	74.1	56.2	−4.3	−2.6	2.9	50.8
	o3	79.9	62.1	−18.1	1.4	1.8	61.6
	Gem. 2.5 Flash	75.9	1.5	45.3	−7.8	25.6	18.9
	Gem. 2.5 Pro	75.9	16.3	22.1	10.9	3.0	50.2
	Cl. Sonnet 4	72.6	12.3	7.8	2.1	43.0	6.9
	Cl. Opus 4	130.4	6.9	22.6	−1.8	−5.7	125.8

C EXTENDED Q1 PORTFOLIO METRICS

Table 7 reports gross full-window performance metrics for the top-quintile (Q1) portfolio across all three prediction modes, including annualized volatility, Calmar ratio, and hit ratio.

Table 7: Extended gross full-window Q1 portfolio metrics, before transaction costs. Bold indicates the best value in each metric column within each panel.

Mode	Model	CAGR	Vol	Sharpe	Sortino	MDD	Calmar	Hit%
<i>Panel A: Exp. Return</i>								
	GPT-4o	41.1	29.8	1.30	1.98	−35.0	1.172	54.0
	GPT-4o mini	−5.3	23.9	−0.11	−0.16	−43.7	−0.122	50.4
	GPT-4.1	92.5	40.5	1.81	2.84	−34.5	2.685	55.3
	GPT-4.1 mini	124.7	32.7	2.62	4.32	−20.8	5.994	56.8
	GPT-4.1 nano	77.9	29.9	2.06	3.35	−21.0	3.707	54.4
	o4-mini	77.9	35.8	1.78	2.78	−30.3	2.573	55.6
	o3	87.0	40.9	1.72	2.64	−31.9	2.727	57.1
	Gem. 2.5 Flash	83.0	38.0	1.77	2.72	−36.7	2.262	57.5
	Gem. 2.5 Pro	62.0	41.7	1.36	2.00	−43.5	1.425	57.0
	Cl. Sonnet 4	145.0	33.8	2.80	4.97	−21.6	6.706	57.8
	Cl. Opus 4	106.7	37.5	2.11	3.32	−33.9	3.152	58.8
<i>Panel B: Sharpe</i>								
	GPT-4o	−6.2	28.8	−0.08	−0.11	−55.2	−0.112	51.1
	GPT-4o mini	−7.0	25.0	−0.17	−0.24	−47.2	−0.149	51.6
	GPT-4.1	124.5	33.3	2.58	4.20	−22.7	5.479	57.1
	GPT-4.1 mini	137.5	37.6	2.47	4.02	−27.7	4.972	57.8
	GPT-4.1 nano	50.1	26.5	1.65	2.55	−23.7	2.115	56.2
	o4-mini	73.5	25.8	2.25	3.43	−24.4	3.011	58.7
	o3	102.5	25.6	2.86	4.76	−10.7	9.574	59.0
	Gem. 2.5 Flash	39.8	30.2	1.25	1.77	−36.7	1.084	55.2
	Gem. 2.5 Pro	85.4	25.2	2.55	4.29	−12.6	6.780	58.5
	Cl. Sonnet 4	104.0	26.9	2.76	4.44	−21.2	4.896	58.5
	Cl. Opus 4	66.4	25.3	2.12	3.68	−18.3	3.620	52.4
<i>Panel C: Sortino</i>								
	GPT-4o	2.0	28.2	0.21	0.31	−33.1	0.062	51.2
	GPT-4o mini	−9.2	23.1	−0.30	−0.42	−45.7	−0.202	49.2
	GPT-4.1	93.9	33.0	2.16	3.42	−23.9	3.923	55.3
	GPT-4.1 mini	60.8	31.2	1.66	2.54	−25.1	2.423	54.4
	GPT-4.1 nano	52.7	25.3	1.79	2.73	−23.6	2.231	55.9
	o4-mini	74.1	27.0	2.18	3.31	−23.9	3.104	58.4
	o3	79.9	26.0	2.38	3.66	−19.8	4.043	59.6
	Gem. 2.5 Flash	75.9	23.9	2.47	3.91	−17.8	4.263	57.5
	Gem. 2.5 Pro	75.9	19.9	2.92	4.80	−9.6	7.935	56.7
	Cl. Sonnet 4	72.6	24.7	2.32	3.74	−20.5	3.535	59.3
	Cl. Opus 4	130.4	30.0	2.91	5.18	−21.9	5.964	57.8

D COMMON-WINDOW AND COST ROBUSTNESS

The main text emphasizes conservative model-specific windows because each LLM has a different public release or documented update boundary. As a complementary check, Table 8 first applies those model-specific conservative boundaries, then restricts LLMs and non-LLM baselines to their common ranking window so the comparison does not rely on different evaluation months or pre-release model histories. This table is not a replacement for the conservative-window results, but it verifies that the strongest LLM rankers remain competitive when the calendar is held fixed.

Table 8: Common-calendar-window long-short performance. All rankers are restricted to the same 2025-07–2025-12 conservative common ranking window with a 2-day execution lag and corrected 40 bps one-way costs on both long and short legs. CAGR, MDD, and IC are reported in percent; bold indicates the best performance value in each column.

Type	Ranker	LS CAGR	LS Shp	LS Sort	MDD	IC	IC Mo.	Avg N	Avg LS Turn
Baseline	Random Forest	-3.7	-0.07	-0.09	-19.7	16.6	6	27.0	2.57
Baseline	Ridge	35.5	1.40	2.11	-14.6	16.7	6	27.0	2.40
Baseline	12–1 Momentum	13.7	0.57	0.84	-18.5	3.4	6	29.0	1.24
Baseline	ROE Quality	-11.3	-0.57	-0.74	-13.7	-2.6	6	27.0	0.29
Baseline	Value (E/P)	-35.6	-1.39	-1.78	-29.8	-4.3	6	27.0	0.13
Baseline	Composite	-62.6	-2.87	-3.51	-41.2	-4.1	6	29.0	1.81
LLM	Claude Sonnet 4	231.4	3.89	7.12	-9.9	48.0	6	28.3	1.84
LLM	Claude Opus 4	536.8	5.99	11.59	-8.5	70.1	6	28.8	1.87
LLM	Gemini 2.5 Flash	380.7	4.79	8.81	-14.1	51.3	6	28.5	2.20
LLM	Gemini 2.5 Pro	458.8	4.99	9.15	-11.4	57.7	6	28.3	1.61
LLM	GPT-4o	-22.3	-0.52	-0.71	-33.1	12.7	6	28.8	2.91
LLM	GPT-4o mini	-70.8	-4.69	-5.23	-50.1	-34.1	6	23.8	2.56
LLM	GPT-4.1	349.6	4.46	7.92	-11.8	45.6	6	28.2	1.57
LLM	GPT-4.1 mini	487.2	5.52	11.21	-8.4	58.2	6	26.3	2.52
LLM	GPT-4.1 nano	123.8	3.11	5.25	-8.4	31.5	6	21.8	2.24
LLM	o4-mini	302.9	4.31	7.68	-11.4	42.0	6	28.5	1.48
LLM	o3	439.7	5.24	9.85	-8.7	49.5	6	29.0	1.92

Table 9 varies one-way transaction costs for all expected-return rankers in the same conservative common window. The long-short Sharpe ranking is stable from zero to 40 bps, while very high costs predictably compress all active strategies. This supports the main-text interpretation that model differences are not solely an artifact of the baseline 40 bps cost assumption.

Table 9: Common-window transaction-cost sensitivity for all expected-return rankers in the 2025-07–2025-12 conservative common ranking window. The long–short portfolio subtracts one-way costs on both long and short turnover; bold indicates the best Sharpe at each cost level.

Ranker	Avg LS Turn	0 bps	20 bps	40 bps	100 bps
Random Forest	2.57	0.43	0.18	-0.07	-0.78
Ridge	2.40	1.89	1.65	1.40	0.70
12–1 Momentum	1.24	0.77	0.67	0.57	0.27
ROE Quality	0.29	-0.43	-0.50	-0.57	-0.77
Value (E/P)	0.13	-1.32	-1.35	-1.39	-1.48
Composite	1.81	-2.64	-2.76	-2.87	-3.18
Claude Sonnet 4	1.84	4.19	4.04	3.89	3.41
Claude Opus 4	1.87	6.29	6.14	5.99	5.49
Gemini 2.5 Flash	2.20	5.10	4.95	4.79	4.27
Gemini 2.5 Pro	1.61	5.22	5.11	4.99	4.61
GPT-4o	2.91	-0.19	-0.35	-0.52	-1.01
GPT-4o mini	2.56	-4.26	-4.49	-4.69	-5.19
GPT-4.1	1.57	4.67	4.57	4.46	4.11
GPT-4.1 mini	2.52	5.87	5.70	5.52	4.90
GPT-4.1 nano	2.24	3.48	3.30	3.11	2.53
o4-mini	1.48	4.53	4.42	4.31	3.95
o3	1.92	5.52	5.39	5.24	4.78

E INPUT ABLATION STUDY

Table 10 reports an input ablation that re-runs each model–objective pair with fundamentals only, market summaries only, or both information sources. From a factor-prior perspective, the ablation asks which information channel sustains alpha and factor loadings rather than which channel maximizes raw CAGR alone. The appendix table reports the measured ablation panel: full-input and ablated-input results are available for all 11 included models and all three objectives. The performance results show that market-only prompts often retain substantial ranking power, especially for stronger frontier models, while fundamentals-only prompts are generally weaker but not uniformly uninformative. The factor-loading extension in Appendix G shows that this performance gap is mirrored in FF6 loadings and style fingerprints: market summaries preserve the growth–momentum factor-like prior, whereas fundamentals-only inputs sharply reduce alpha and make style fingerprints less economically coherent.

Table 10: Input ablation results by model and objective. CAGR and IC are reported in percent; Sharpe is annualized; bold indicates the best input setting for each metric within each model-objective row.

Objective	Model	Fund. + Market			Fund. only			Market only		
		CAGR	Shp	IC	CAGR	Shp	IC	CAGR	Shp	IC
Exp. Return	GPT-4o	-20.6	-0.58	3.6	0.3	0.15	2.4	217.7	3.79	50.1
Exp. Return	GPT-4o mini	-43.2	-2.30	-20.5	10.4	0.59	-1.2	-64.2	-3.73	-35.6
Exp. Return	GPT-4.1	87.0	1.94	27.1	16.5	0.64	10.4	196.9	3.17	32.7
Exp. Return	GPT-4.1 mini	109.2	2.27	29.4	39.1	1.07	12.4	-27.8	-0.85	1.4
Exp. Return	GPT-4.1 nano	21.0	0.83	16.2	-16.7	-0.67	0.5	58.6	1.82	18.2
Exp. Return	o4-mini	49.6	1.34	24.6	26.0	0.90	5.8	84.6	1.99	20.1
Exp. Return	o3	132.7	2.52	31.0	43.0	1.18	9.6	158.4	2.83	30.4
Exp. Return	Gem. 2.5 Flash	100.8	2.03	34.4	1.3	0.25	-6.7	61.1	1.19	47.5
Exp. Return	Gem. 2.5 Pro	93.2	1.81	27.2	22.1	0.76	8.5	107.2	1.91	23.4
Exp. Return	Cl. Sonnet 4	133.5	2.77	37.6	15.3	0.61	4.5	145.2	2.92	37.0
Exp. Return	Cl. Opus 4	129.1	2.48	43.3	15.2	0.60	4.7	141.4	2.61	37.1
Sharpe	GPT-4o	-59.2	-2.60	-24.8	-7.8	-0.19	0.6	107.1	2.39	31.1
Sharpe	GPT-4o mini	-43.3	-2.30	-15.5	-1.9	0.02	-1.5	-46.9	-2.06	-16.9
Sharpe	GPT-4.1	95.9	2.12	32.6	-2.3	0.06	5.5	103.5	2.44	31.7
Sharpe	GPT-4.1 mini	138.9	2.81	34.6	53.8	1.14	12.9	33.9	1.09	18.3
Sharpe	GPT-4.1 nano	8.0	0.42	18.8	-21.0	-0.95	4.0	-8.2	-0.23	13.3
Sharpe	o4-mini	30.3	0.89	24.0	-6.3	-0.10	3.1	4.4	0.30	18.3
Sharpe	o3	81.1	2.01	30.6	-24.9	-1.10	1.3	67.9	1.87	28.6
Sharpe	Gem. 2.5 Flash	-1.7	0.17	19.1	14.2	0.56	2.0	250.7	3.19	-10.0
Sharpe	Gem. 2.5 Pro	14.7	0.54	25.0	-12.3	-0.42	0.2	83.1	1.83	20.1
Sharpe	Cl. Sonnet 4	121.9	2.43	32.8	-25.2	-0.98	-1.6	81.6	2.09	31.4
Sharpe	Cl. Opus 4	33.8	1.01	23.0	-22.9	-0.85	0.8	103.8	2.58	30.0
Sortino	GPT-4o	-52.3	-2.25	-15.6	-0.8	0.08	1.5	-37.5	-1.31	-10.7
Sortino	GPT-4o mini	-49.3	-2.84	-23.6	-0.4	0.08	1.9	-52.7	-2.49	-18.2
Sortino	GPT-4.1	34.8	1.02	19.5	2.8	0.24	12.7	151.8	2.93	37.6
Sortino	GPT-4.1 mini	71.5	1.88	35.9	22.0	0.64	-2.1	-24.4	-0.83	-0.7
Sortino	GPT-4.1 nano	-9.5	-0.24	11.8	8.7	0.48	9.6	66.8	2.16	20.1
Sortino	o4-mini	39.2	1.06	24.2	-29.2	-1.31	-0.0	37.0	1.26	19.3
Sortino	o3	46.4	1.26	31.5	-32.5	-1.51	0.6	88.5	2.28	28.9
Sortino	Gem. 2.5 Flash	9.2	0.43	19.8	5.3	0.33	17.5	69.6	1.52	-7.5
Sortino	Gem. 2.5 Pro	37.4	1.01	29.1	1.5	0.19	1.0	82.1	1.88	26.1
Sortino	Cl. Sonnet 4	-4.2	0.09	28.5	-36.0	-1.67	-6.2	110.4	2.70	35.3
Sortino	Cl. Opus 4	102.7	2.12	38.2	-11.3	-0.33	-0.6	178.6	3.36	40.1

To formally compare information sets, Table 11 tests paired differences between the full input and each ablated input. Each observation is a matched model-objective cell, and the test flips the sign of all three objectives within each model cluster, preserving within-model

dependence while testing whether the average paired difference is zero. The full input is not significantly different from market-only prompts on long-short CAGR, Sharpe, or mean IC, indicating that the market summary block carries most of the measured ranking signal in this sample. By contrast, the full input significantly improves long-short CAGR and mean IC relative to fundamentals-only prompts, with a positive but less precisely estimated Sharpe difference.

Table 11: Paired input-ablation tests. Differences are full input minus the ablated input; p-values use model-clustered sign-flip tests.

Comparison	Metric	Both	Ablated	Δ	95% CI	p
Both - Market only	LS CAGR	43.6	76.7	-33.1	[-67.8, 5.0]	0.119
Both - Market only	LS Sharpe	0.79	1.41	-0.62	[-1.40, 0.19]	0.183
Both - Market only	Mean IC	19.8	19.0	0.8	[-8.4, 8.9]	0.857
Both - Fund. only	LS CAGR	43.6	1.4	42.2	[12.4, 69.2]	0.026
Both - Fund. only	LS Sharpe	0.79	0.02	0.78	[-0.23, 1.62]	0.147
Both - Fund. only	Mean IC	19.8	3.5	16.3	[5.6, 25.3]	0.018

F BENCHMARK IMPLICATIONS

The results suggest three rules for future cross-sectional LLM ranker benchmarks. First, baselines should include factor sorts and lightweight supervised rankers. Factor sorts test whether an LLM is rediscovering familiar style premia, while expanding-window models test whether the same panel already contains enough signal for a conventional learner. The gap between Random Forest/Ridge and the strongest LLMs in Tables 2–4 is therefore more informative than a comparison against the market index alone.

Second, evaluation should keep the ranking object fixed. Pointwise scores, pairwise tournaments, setwise reranking, and agentic debate can all be useful, but they should ultimately output the same monthly ordering over the same investable universe. This makes IC, quintile monotonicity, long–short spreads, and factor diagnostics directly comparable across methods. It also prevents an agentic system from changing the task by using additional data, extra lookups, or different rebalancing assumptions.

Third, cost and reproducibility should be first-class metrics. Listwise prompting is attractive because it obtains the full cross-section in one call, whereas pairwise and agentic systems can scale call counts quickly. Reporting cost per monthly experiment, malformed-output handling, decoding settings, and conservative evaluation boundaries makes it possible to distinguish genuine ranking skill from engineering choices that are expensive or difficult to replicate.

These rules also clarify how to read the negative findings. The underperforming models are not simply lower-return versions of the winners; they encode different implicit priors about the cross-section. GPT-4o mini, for example, ranks toward reversal and high-volatility names in a period when the opposite style was rewarded. A future benchmark should therefore measure not only whether a model wins in one sample but also whether its style exposure is stable, interpretable, and robust when market leadership changes.

Finally, the current evidence should be viewed as a controlled starting point rather than a finished trading system. The 30-stock universe is small enough for transparent listwise prompting, and the transaction-cost model is intentionally simple. Broader universes, rolling sector composition, liquidity constraints, and turnover-aware construction are natural next steps. Future work should add broader universes, rolling membership, and richer information axes—such as disclosure text, which recent real-time event benchmarks show can explain announcement returns [26]—while keeping raw ranking quality separate from weighting or optimization gains.

G FACTOR LOADINGS UNDER INPUT ABLATIONS

The factor-prior interpretation in Section 4.3 asks not only whether LLM rankings have alpha, but *which information channel* generates the implicit ranking profile.

Table 12 reports FF6 loadings and alphas under fundamentals-only, market-only, and full-input settings for o3, Gemini 2.5 Pro, and Claude Opus 4. Table 13 reports the corresponding within-universe style fingerprint for the same representative models. Figure 6 visualizes how factor exposures shift when fundamentals or market summaries are removed.

Two patterns emerge. First, **market-only inputs preserve most of the full-input factor profile**: positive momentum and negative value/profitability/investment loadings remain for the strongest models, consistent with the hypothesis-test result that full-input and market-only performance are not significantly different. Second, **fundamentals-only inputs weaken both alpha and style coherence**: some FF6 market and momentum exposure remains, but alpha collapses and characteristic fingerprints become noisier, indicating that sparse accounting panels alone do not sustain a stable economically useful factor-like prior in this universe.

Table 12: FF6 factor loadings of representative Q1–Q5 portfolios under input ablations. α is annualized percent; stars use Newey–West HAC SE.

Model	Input	α	β_{MKT}	β_{SMB}	β_{HML}	β_{RMW}	β_{CMA}	β_{MOM}	R^2_{adj}
o3	Fund. + Market	127.7	+0.86***	-0.06	-0.70***	-0.60**	+0.28	+0.34**	0.45
o3	Market only	156.5	+0.76***	+0.20	-0.81***	-0.36	+0.38	+0.68***	0.48
o3	Fund. only	39.7	+0.68***	+0.05	-0.82***	-0.36	+0.24	+0.67***	0.44
Gem. 2.5 Pro	Fund. + Market	87.6	+0.83***	-0.06	-0.64***	-0.75***	+0.40	+0.51***	0.38
Gem. 2.5 Pro	Market only	116.8	+1.02***	-0.43	-0.86***	-0.43	+0.85***	+0.32	0.42
Gem. 2.5 Pro	Fund. only	24.9	+0.50***	-0.00	-0.97***	-0.30	+0.48**	+0.66***	0.44
Cl. Opus 4	Fund. + Market	129.6	+0.68***	-0.06	-0.18	-0.42	-0.06	+0.59***	0.26
Cl. Opus 4	Market only	143.6	+0.58***	+0.16	-0.78***	-0.44*	+0.97***	+0.79***	0.44
Cl. Opus 4	Fund. only	12.2	+0.55***	-0.07	-0.96***	-0.32*	+0.26	+0.76***	0.56

Table 13: Within-universe style fingerprint under input ablations. Values are average Q1–Q5 normalized characteristic ranks; stars test the monthly mean spread with Newey–West HAC SE: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Model	Input	Mom	Rev	LowVol	E/P	ROE	LowInv
o3	Fund. + Market	+0.04	-0.21*	-0.57***	-0.21***	-0.05***	-0.10***
o3	Market only	+0.17**	-0.23**	-0.48***	-0.33***	-0.15***	-0.20***
o3	Fund. only	+0.15	-0.13	-0.47***	-0.37***	-0.06**	-0.21***
Gem. 2.5 Pro	Fund. + Market	+0.24***	-0.22**	-0.52***	-0.41***	-0.29***	-0.12**
Gem. 2.5 Pro	Market only	+0.23**	-0.10	-0.38***	-0.30**	-0.05**	-0.15***
Gem. 2.5 Pro	Fund. only	+0.21	-0.04	-0.37***	-0.43***	+0.01	-0.23***
Cl. Opus 4	Fund. + Market	+0.32***	-0.16***	-0.35***	-0.31***	-0.13*	-0.23***
Cl. Opus 4	Market only	+0.23**	-0.13*	-0.43***	-0.36***	-0.04**	-0.15***
Cl. Opus 4	Fund. only	+0.40***	-0.11**	-0.38***	-0.46***	+0.10**	-0.32***

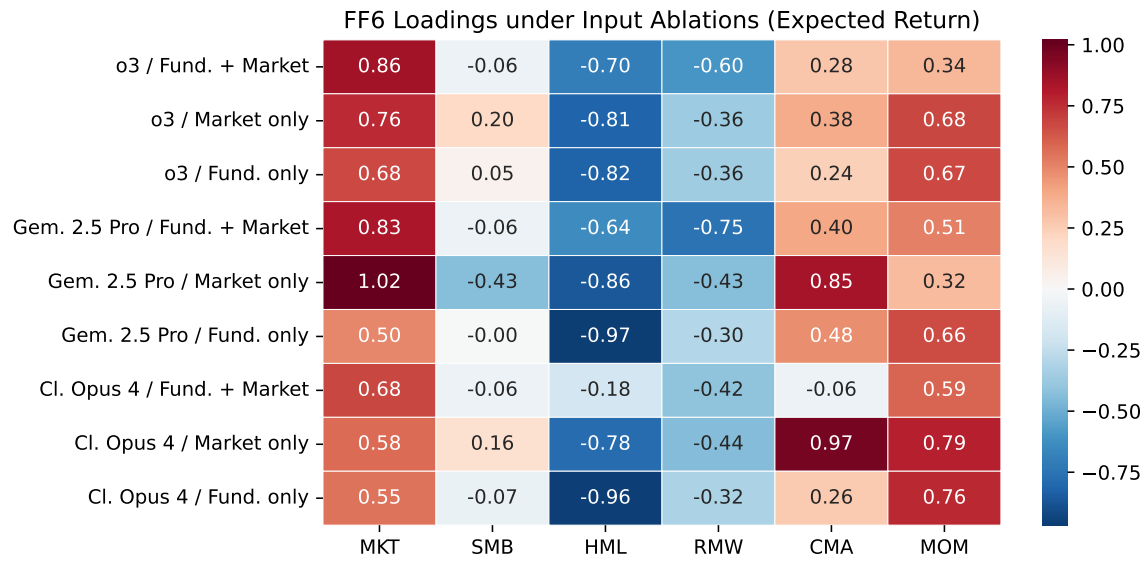


Figure 6: FF6 factor loadings of representative Q1–Q5 portfolios under input ablations. Red indicates positive loading and blue indicates negative loading.

H LONG-ONLY Q1 CUMULATIVE RETURNS

Figure 7 shows the cumulative wealth path of the top-quintile (Q1) long-only portfolio for each model and prediction mode, displayed from 2025-05 after compounding each path from its conservative start. These trajectories combine the ranking signal with broad market exposure and are therefore more sensitive to market direction than the Q1–Q5 spread in the main text; the equal-weighted universe line helps separate stock-selection gains from the common AI-rally market drift.

Q1 (Top Quintile) Cumulative Returns (Appendix)

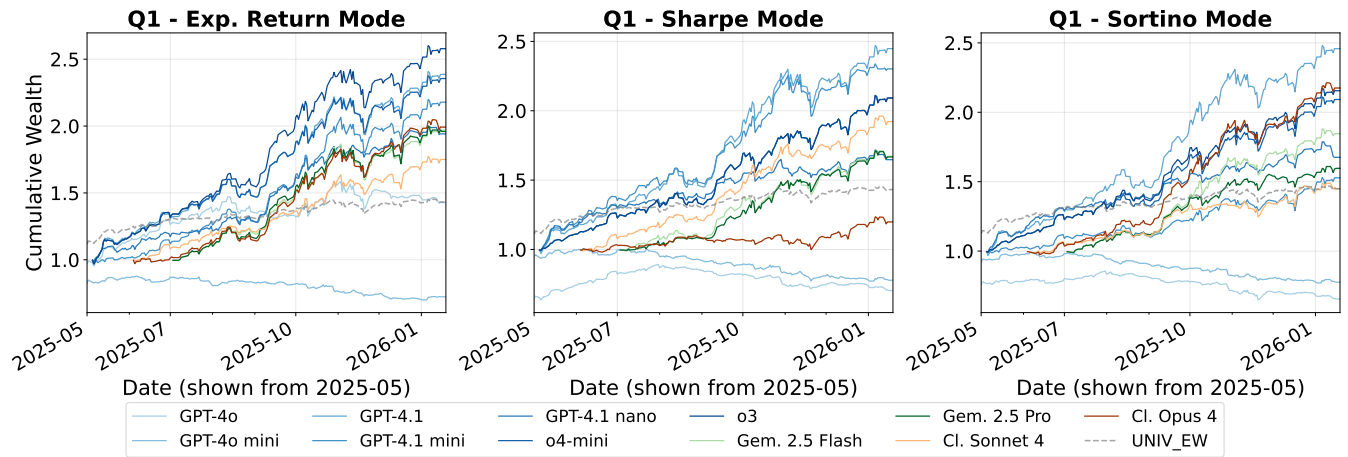


Figure 7: Cumulative wealth of Q1 long-only portfolios by model and prediction mode, shown from 2025-05 after compounding each path from its conservative start. The dashed gray line is the equal-weighted universe benchmark.

The Q1 paths reinforce why the main text emphasizes Q1–Q5 spreads rather than long-only returns alone. Several strong models participate in the broad AI-market rally, but the market-neutral spread tests whether the same ranking also avoids the weak bottom quintile. Thus, Figure 7 is best read as a long-only decomposition of the stronger cross-sectional evidence in Table 2.

I LONG-ONLY Q1 PERFORMANCE HEATMAP

Figure 8 shows the top-quintile (Q1) long-only performance heatmap, complementing the long-short summary in Table 2. It is useful for separating models that simply ride the broad market in Q1 from models whose bottom-quintile avoidance creates a stronger spread.

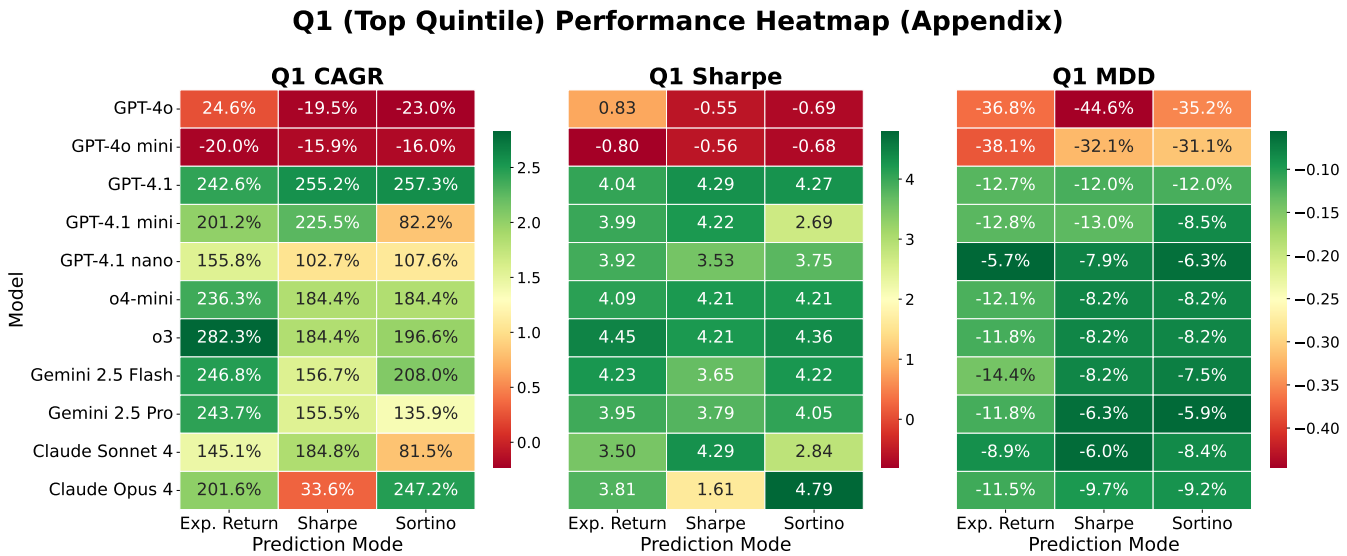


Figure 8: Performance heatmap of Q1 long-only portfolios by model and prediction mode. Green indicates stronger values; MDD is inverted.

The heatmap shows that high Q1 returns are not identical to robust ranking skill. Models can look attractive on long-only CAGR while still producing weaker market-neutral separation if their bottom quintile also benefits from the same market regime. This is why the main benchmark reports Q1 metrics, Q1–Q5 spreads, and IC together.

J LONG-SHORT PERFORMANCE HEATMAP

Figure 9 summarizes the long-short Q1–Q5 portfolio metrics across model-objective pairs, complementing the compact main-text Table 2 and the extended metrics in Table 14. Compared with the Q1 heatmap, this view emphasizes whether the full ranking separates winners from losers rather than merely selecting a strong top basket.

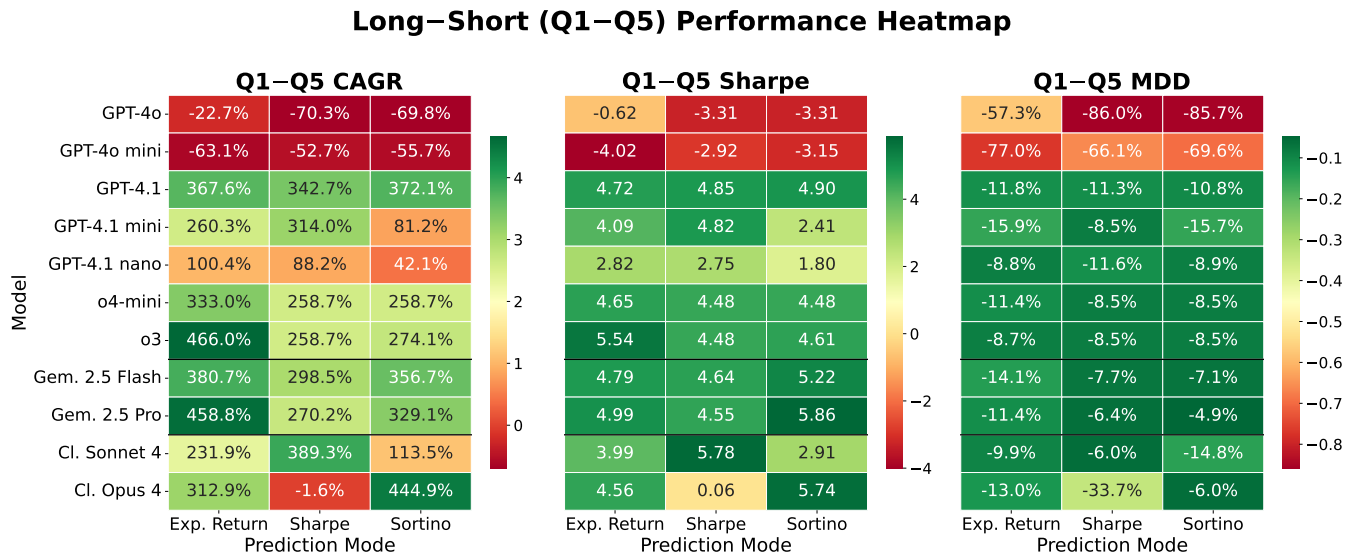


Figure 9: Performance heatmap of Q1–Q5 long-short portfolios by model and prediction mode. Green indicates stronger values; MDD is inverted.

This view is the closest appendix counterpart to the main economic test. The strongest cells combine high spread returns with manageable drawdowns, while the weakest models show that an LLM can be systematically wrong rather than merely noisy. The contrast with Figure 8 helps separate market exposure from true cross-sectional ordering.

K EXTENDED LONG–SHORT PORTFOLIO METRICS

Table 14 reports gross full-window extended performance metrics for the long–short (Q1–Q5) portfolio, complementing the net 40 bps summary in Table 2.

Table 14: Extended gross full-window Q1–Q5 long–short metrics, before transaction costs. Bold indicates the best value in each metric column within each panel.

Mode	Model	CAGR	Vol	Sharpe	Sortino	MDD	Calmar	Hit%
<i>Panel A: Exp. Return</i>								
	GPT-4o	−9.3	31.2	−0.16	−0.21	−59.7	−0.156	48.8
	GPT-4o mini	−37.0	23.2	−1.87	−2.45	−79.5	−0.465	43.1
	GPT-4.1	105.1	35.0	2.21	3.48	−31.2	3.375	54.1
	GPT-4.1 mini	138.4	34.6	2.67	4.45	−32.8	4.224	52.3
	GPT-4.1 nano	34.6	27.2	1.22	1.82	−23.8	1.452	52.0
	o4-mini	64.9	34.2	1.63	2.38	−43.7	1.486	54.1
	o3	153.2	35.7	2.76	4.44	−24.9	6.156	56.5
	Gem. 2.5 Flash	120.7	37.9	2.26	3.49	−36.0	3.352	57.5
	Gem. 2.5 Pro	107.9	40.8	1.99	2.97	−37.3	2.893	57.3
	Cl. Sonnet 4	158.2	32.4	3.07	5.60	−18.5	8.542	55.5
	Cl. Opus 4	156.3	35.7	2.80	4.48	−33.9	4.615	59.5
<i>Panel B: Sharpe</i>								
	GPT-4o	−53.5	32.1	−2.22	−2.75	−91.3	−0.586	43.8
	GPT-4o mini	−37.4	23.2	−1.90	−2.53	−77.9	−0.480	42.4
	GPT-4.1	114.9	34.0	2.41	3.80	−30.4	3.775	53.5
	GPT-4.1 mini	168.3	32.3	3.20	5.65	−17.4	9.695	57.4
	GPT-4.1 nano	20.5	26.7	0.83	1.13	−25.9	0.789	54.7
	o4-mini	40.8	37.5	1.10	1.55	−42.7	0.956	52.6
	o3	96.6	32.0	2.26	3.46	−25.9	3.727	56.2
	Gem. 2.5 Flash	8.3	41.4	0.40	0.54	−65.9	0.127	55.2
	Gem. 2.5 Pro	27.0	41.6	0.79	1.03	−63.1	0.427	53.2
	Cl. Sonnet 4	147.3	34.9	2.75	4.13	−32.6	4.516	57.5
	Cl. Opus 4	53.5	34.3	1.41	2.12	−32.0	1.670	51.7
<i>Panel C: Sortino</i>								
	GPT-4o	−45.4	30.3	−1.84	−2.34	−85.5	−0.531	44.7
	GPT-4o mini	−43.8	22.7	−2.42	−3.11	−82.4	−0.532	40.7
	GPT-4.1	49.7	35.2	1.31	1.95	−49.5	1.004	50.2
	GPT-4.1 mini	95.3	30.6	2.33	3.95	−24.2	3.945	53.2
	GPT-4.1 nano	−1.2	26.3	0.09	0.12	−34.7	−0.034	52.3
	o4-mini	50.8	37.7	1.27	1.76	−43.2	1.175	54.7
	o3	61.6	34.7	1.55	2.18	−32.4	1.901	54.4
	Gem. 2.5 Flash	18.9	43.7	0.62	0.80	−65.0	0.291	52.7
	Gem. 2.5 Pro	50.2	38.5	1.24	1.76	−56.8	0.883	54.7
	Cl. Sonnet 4	6.9	37.4	0.37	0.47	−62.5	0.111	54.2
	Cl. Opus 4	125.8	35.9	2.43	3.76	−31.9	3.946	55.7

L WITHIN-UNIVERSE FACTOR-MIMICKING PORTFOLIOS

Because the 30-stock universe is concentrated in mega-cap technology, global Fama–French factors may not capture style tilts that occur *within* the universe. For each characteristic c , we construct a monthly within-universe long–short portfolio by sorting the 30 stocks on c , going long the top 6 (on the traditional “long” side of that characteristic) and short the bottom 6, rebalancing with a 2-day lag and no transaction cost. Table 15 reports the daily-return Pearson correlation between each LLM’s Q1–Q5 portfolio and these factor-mimicking portfolios (expected-return mode).

Table 15: Correlation between LLM Q1–Q5 returns and within-universe factor-mimicking portfolios. Positive entries align with the factor’s traditional long side; stars test zero correlation with Newey–West HAC SE: $*p < 0.10$, $**p < 0.05$, $***p < 0.01$.

Model	Mom	Rev	LowVol	E/P	ROE	LowInv	SmallSize
GPT-4o	−0.02	−0.04	−0.18**	−0.18**	−0.10	−0.13*	−0.23***
GPT-4o mini	−0.20***	+0.22***	+0.22***	+0.21***	−0.00	+0.20***	+0.11*
GPT-4.1	+0.37***	−0.38***	−0.70***	−0.57***	−0.31***	−0.47***	−0.29***
GPT-4.1 mini	+0.29***	−0.39***	−0.12	−0.16	−0.02	−0.02	−0.20***
GPT-4.1 nano	+0.13	−0.21**	−0.23**	−0.13	−0.08	−0.08	−0.18**
o4-mini	+0.24**	−0.11	−0.65***	−0.46***	−0.29***	−0.28***	−0.24***
o3	+0.28***	−0.15	−0.80***	−0.60***	−0.35***	−0.49***	−0.29***
Gem. 2.5 Flash	+0.48***	−0.32***	−0.54***	−0.57***	−0.21***	−0.23**	−0.17**
Gem. 2.5 Pro	+0.44***	−0.13	−0.78***	−0.67***	−0.42***	−0.34***	−0.14**
Cl. Sonnet 4	+0.37***	−0.34***	−0.50***	−0.53***	−0.21***	−0.40***	−0.27***
Cl. Opus 4	+0.51***	−0.23**	−0.59***	−0.61***	−0.24***	−0.42***	−0.20***

The correlation structure is consistent with the FF6 loadings in Table 4 and the style fingerprint in Figure 3: the strongest representative models—o3, Gemini 2.5 Pro, and Claude Opus 4—all show positive momentum correlation and negative low-volatility/value correlations, confirming a high-momentum, high-volatility growth tilt within the universe. GPT-4o mini remains the main counterexample, with negative momentum and positive reversal/low-volatility correlations, consistent with its anti-monotonic quintile pattern during the 2024–2025 AI rally.

Figure 10 visualizes the FF6 factor loadings as a heatmap, offering a compact view of the cross-vendor style differences.

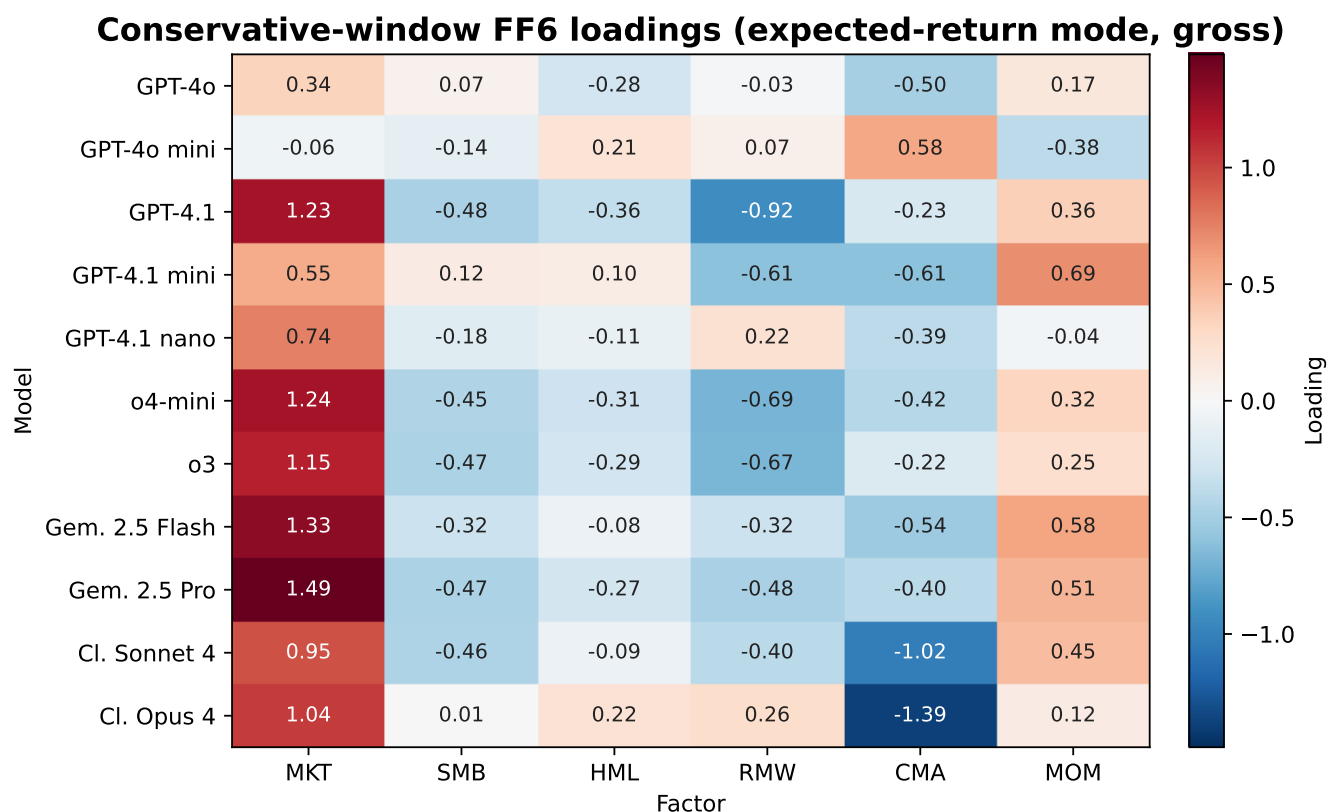


Figure 10: Conservative-window FF6 factor loadings of expected-return Q1–Q5 portfolios. Red indicates positive loading and blue indicates negative loading.

The heatmap closes the appendix by linking the table-level style correlations back to the FF6 spanning tests. Across the stronger expected-return rankers, positive market and momentum exposure coexists with negative value/profitability/investment loadings, consistent with the growth–momentum implicit-prior interpretation developed in the main text.